

Book Club Architecture Justification

Problem Statement 55

Book Club Reading Challenge and Discussion Portal

Architecture selection

Layered architecture is selected for the Book Club Reading Challenge and Discussion Portal. Presentation, Business, and Data layers separate the browser interface, club rules, and persistent records. The Business components form one modular server application.

Reason 1 Separate club rules

Reading goals, spoiler visibility, and monthly polls have different validation rules. Dedicated Reading Challenge, Discussion, and Poll components keep these rules separate. A change to spoiler handling does not require rewriting vote logic, and each component can be tested through its interface.

Reason 2 Consistent shared records

The features share member identity and club data. A common Repository Gateway and relational database keep access consistent. Recording a vote and updating its tally in one transaction, with a unique member-and-poll constraint, prevents duplicate votes even when requests arrive together.

Security advantage

Application API verifies active sessions, club membership, record ownership, and role permissions before dispatching protected operations. The browser has no direct database access. Discussion Service removes hidden spoiler text from previews and search responses until reveal is permitted; Repository Gateway uses parameterized queries.

Performance benefit

In-process calls between Business components avoid inter-service network round trips. Indexed lookups support goal, discussion, and poll queries. Server-Sent Events deliver committed tally changes without repeated full-page polling. These are expected benefits; concurrent-load tests are still needed to measure latency and assess poll-row lock contention.

Tradeoff and availability

Individual modules cannot scale independently, and the application remains a central failure point. Durable commits preserve saved logs and comments across application restarts. Monitoring, automatic restart, backups, and restoration tests support the 99% monthly availability target, which must be verified through uptime measurements.